Decomposition analyses exc. anonomalous savanna sites (AI; reordered; dredge-selected best models refit with REML)

30 April 2026

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0.1 Packages

0.2 Data (CSV, filtering, AI extraction)

Data summary

CSV_used	N	Sites	AI_min	AI_max
C:/Users/RTHOR95/OneDrive - The University of Liverpool/R/joel_termites/Termite_Decomp/Termite_Decomp/Updated_DB4_NEW.csv	37	12	0.353	1.792

0.3 Helpers (print-friendly tables, best-model refit, R², plotting style)

1 1) Termite absolute (Term_Absolute)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
80.86	-33.15	NA	+	NA	NA	5	-143.64	314.34	0.00	0.40

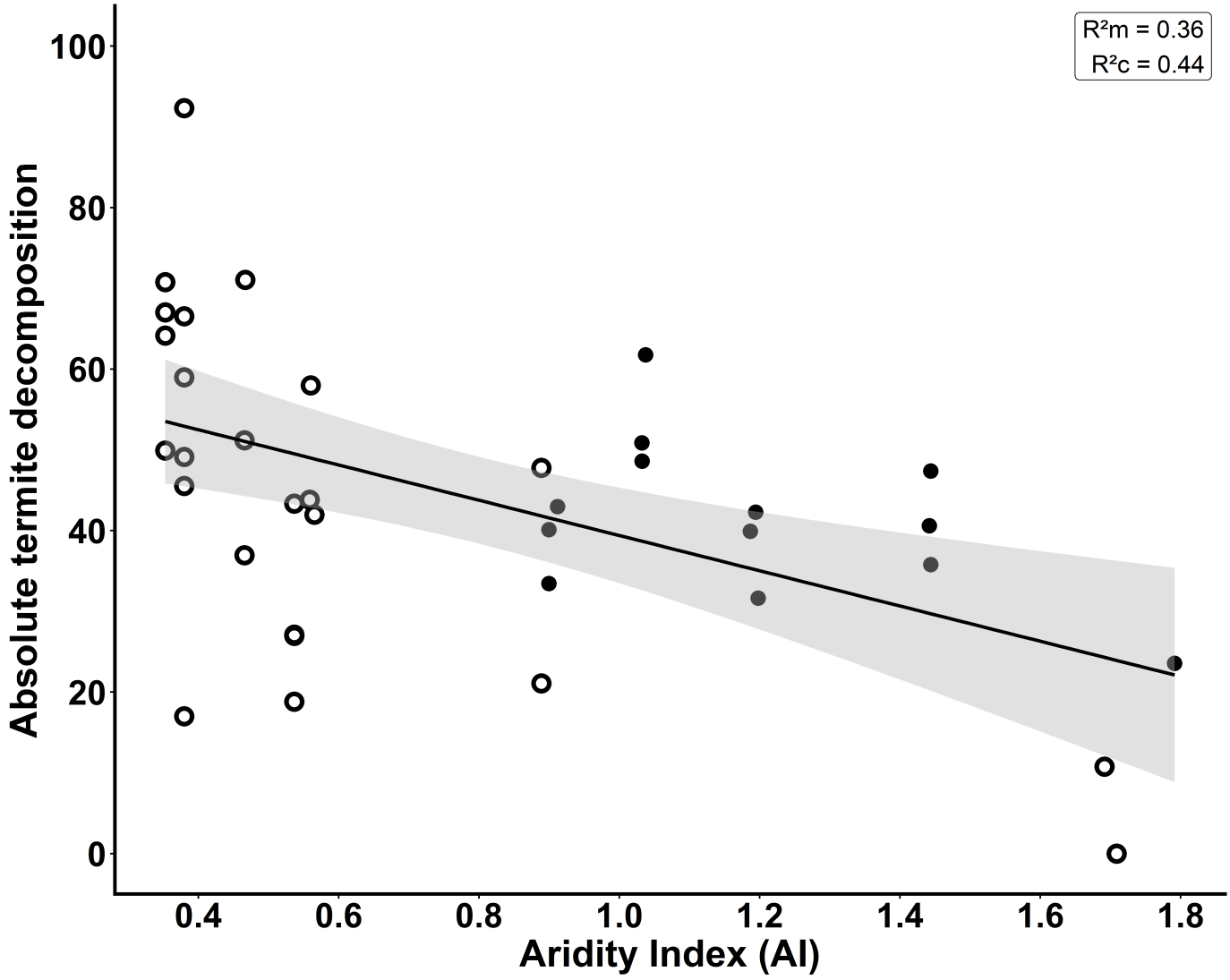
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
61.56	-17.09	NA	+	+	NA	6	-139.20	315.91	1.57	0.18
52.44	-34.90	1.18	+	NA	NA	6	-142.03	316.54	2.20	0.13
62.47	-23.59	NA	NA	NA	NA	4	-148.80	316.90	2.56	0.11
23.68	-16.73	1.45	+	+	NA	7	-137.41	317.76	3.42	0.07
26.58	-26.30	1.53	NA	NA	NA	5	-146.95	318.72	4.38	0.05
178.68	-35.60	-3.66	+	NA	+	7	-138.55	319.56	5.21	0.03
39.24	-17.16	0.87	+	+	+	8	-133.98	321.32	6.97	0.01
40.63	NA	NA	NA	NA	NA	3	-156.14	323.90	9.56	0.00
41.22	NA	NA	+	NA	NA	4	-153.11	326.47	12.13	0.00
53.97	NA	-0.54	NA	NA	NA	4	-154.40	326.53	12.19	0.00
58.16	NA	-0.67	+	NA	NA	5	-151.30	329.29	14.95	0.00
-162.42	NA	7.84	+	NA	+	6	-147.42	332.03	17.69	0.00

Selected_model_formula
Term_Absolute ~ AI + VegType + (1 | Site)

nobs	AIC	BIC	logLik	sigma
37	297.28	305.33	-143.64	14.11

term	estimate	std_error	df	t_value
(Intercept)	80.86	10.33	24.28	7.82
AI	-33.15	7.77	25.65	-4.26
VegTypeSavanna	-16.12	7.13	28.11	-2.26

R2m	R2c
0.36	0.44



2 2) Microbe absolute (Microbe_Absolute)

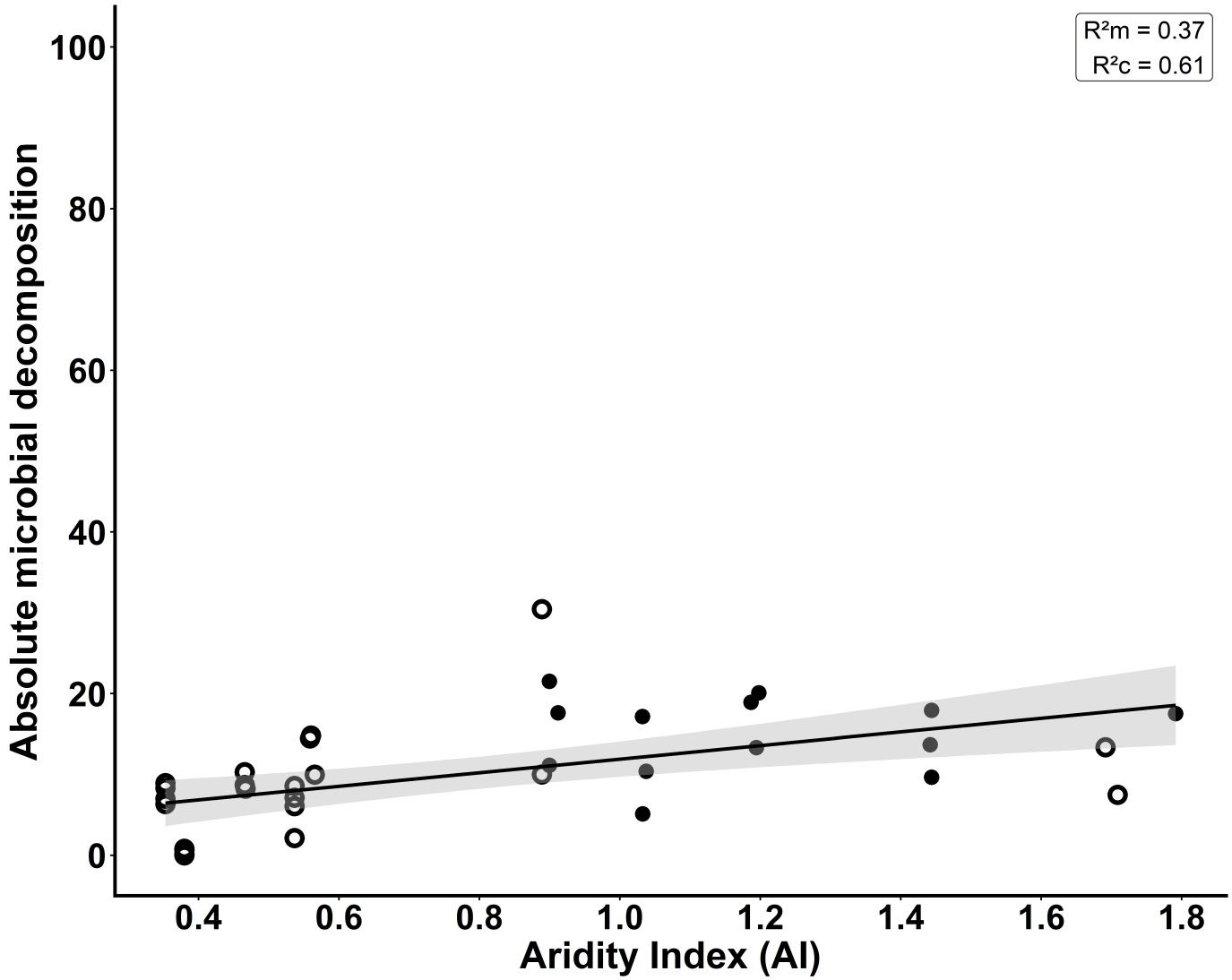
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
-24.04	4.27	1.24	NA	NA	NA	5	-107.21	233.01	0.00	0.24
-28.10	NA	1.56	NA	NA	NA	4	-110.30	233.09	0.09	0.23
-22.35	NA	1.40	+	NA	NA	5	-107.90	233.95	0.94	0.15
5.47	5.94	NA	NA	NA	NA	4	-109.52	234.75	1.74	0.10
-21.64	3.51	1.21	+	NA	NA	6	-105.20	235.31	2.30	0.08
11.02	NA	NA	NA	NA	NA	3	-113.36	236.27	3.26	0.05
56.38	NA	-1.64	+	NA	+	6	-105.28	236.51	3.50	0.04
7.02	5.24	NA	+	NA	NA	5	-107.48	237.04	4.03	0.03
12.99	NA	NA	+	NA	NA	4	-110.89	237.27	4.26	0.03
-27.93	7.34	1.27	+	+	NA	7	-102.07	238.38	5.37	0.02
24.99	3.18	-0.58	+	NA	+	7	-102.71	238.46	5.45	0.02
3.10	8.38	NA	+	+	NA	6	-104.35	239.90	6.90	0.01
-9.11	7.16	0.55	+	+	+	8	-99.49	242.00	9.00	0.00

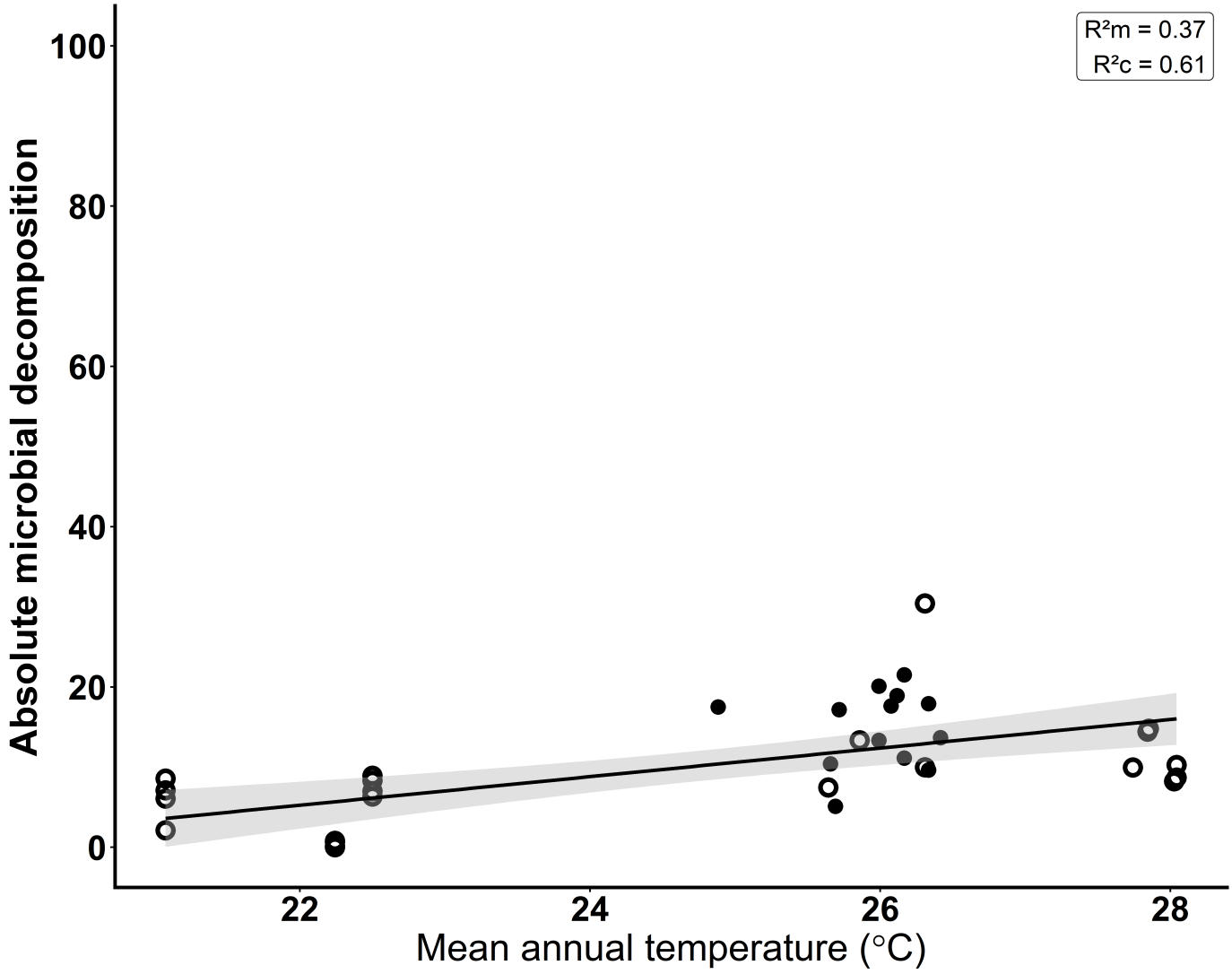
Selected_model_formula
Microbe_Absolute ~ AI + Avg_Temp + (1 | Site)

nobs	AIC	BIC	logLik	sigma
37	224.43	232.48	-107.21	4.23

term	estimate	std_error	df	t_value
(Intercept)	-24.04	14.36	7.95	-1.67
AI	4.27	2.70	13.01	1.58
Avg_Temp	1.24	0.60	8.39	2.07

R2m	R2c
0.37	0.61





3 3) Total decomposition (Term_Loss)

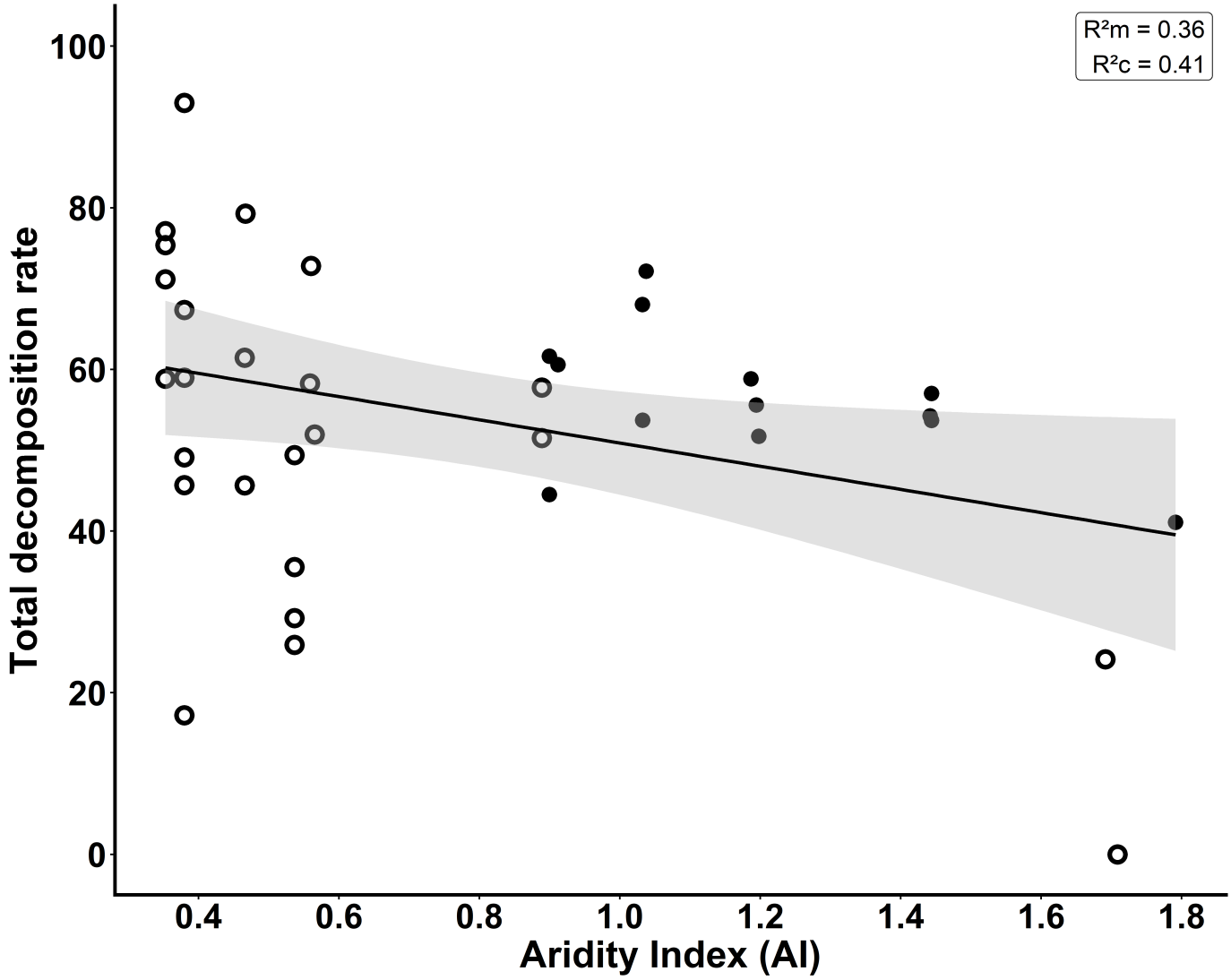
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
34.32	-34.17	2.41	+	NA	NA	6	-142.03	315.89	0.00	0.33
1.12	-13.43	2.73	+	+	NA	7	-137.14	316.56	0.67	0.23
91.25	-29.89	NA	+	NA	NA	5	-144.82	316.94	1.05	0.19
70.09	-12.50	NA	+	+	NA	6	-140.31	318.72	2.83	0.08
189.11	-34.87	-3.52	+	NA	+	7	-138.54	318.86	2.97	0.07
43.89	-13.91	1.10	+	+	+	8	-133.76	320.11	4.22	0.04
69.11	-19.55	NA	NA	NA	NA	4	-150.67	321.37	5.48	0.02
1.65	-23.57	2.84	NA	NA	NA	5	-148.01	321.43	5.54	0.02
51.07	NA	NA	NA	NA	NA	3	-156.20	324.06	8.17	0.01
54.17	NA	NA	+	NA	NA	4	-152.99	326.21	10.32	0.00
26.84	NA	0.95	NA	NA	NA	4	-154.39	326.56	10.67	0.00
36.05	NA	0.70	+	NA	NA	5	-151.18	329.02	13.13	0.00
-85.31	NA	5.37	+	NA	+	6	-147.39	331.99	16.10	0.00

Selected_model_formula
Term_Loss ~ AI + Avg_Temp + VegType + (1 | Site)

nobs	AIC	BIC	logLik	sigma
37	296.05	305.72	-142.03	14.52

term	estimate	std_error	df	t_value
(Intercept)	34.32	32.31	7.79	1.06
AI	-34.17	7.87	24.46	-4.34
Avg_Temp	2.41	1.28	7.48	1.88
VegTypeSavanna	-20.01	7.02	25.69	-2.85

R2m	R2c
0.36	0.41



4 4) Termite proportional (Term_Prop)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
89.78	-14.64	NA	+	+	NA	6	-126.54	289.11	0.00	0.69
99.68	-14.55	-0.40	+	+	NA	7	-125.27	292.56	3.45	0.12
123.61	-43.37	NA	+	NA	NA	5	-132.77	292.57	3.46	0.12
141.33	-42.21	-0.76	+	NA	NA	6	-131.23	295.47	6.36	0.03
14.48	-11.80	2.76	+	+	+	8	-121.86	296.41	7.30	0.02

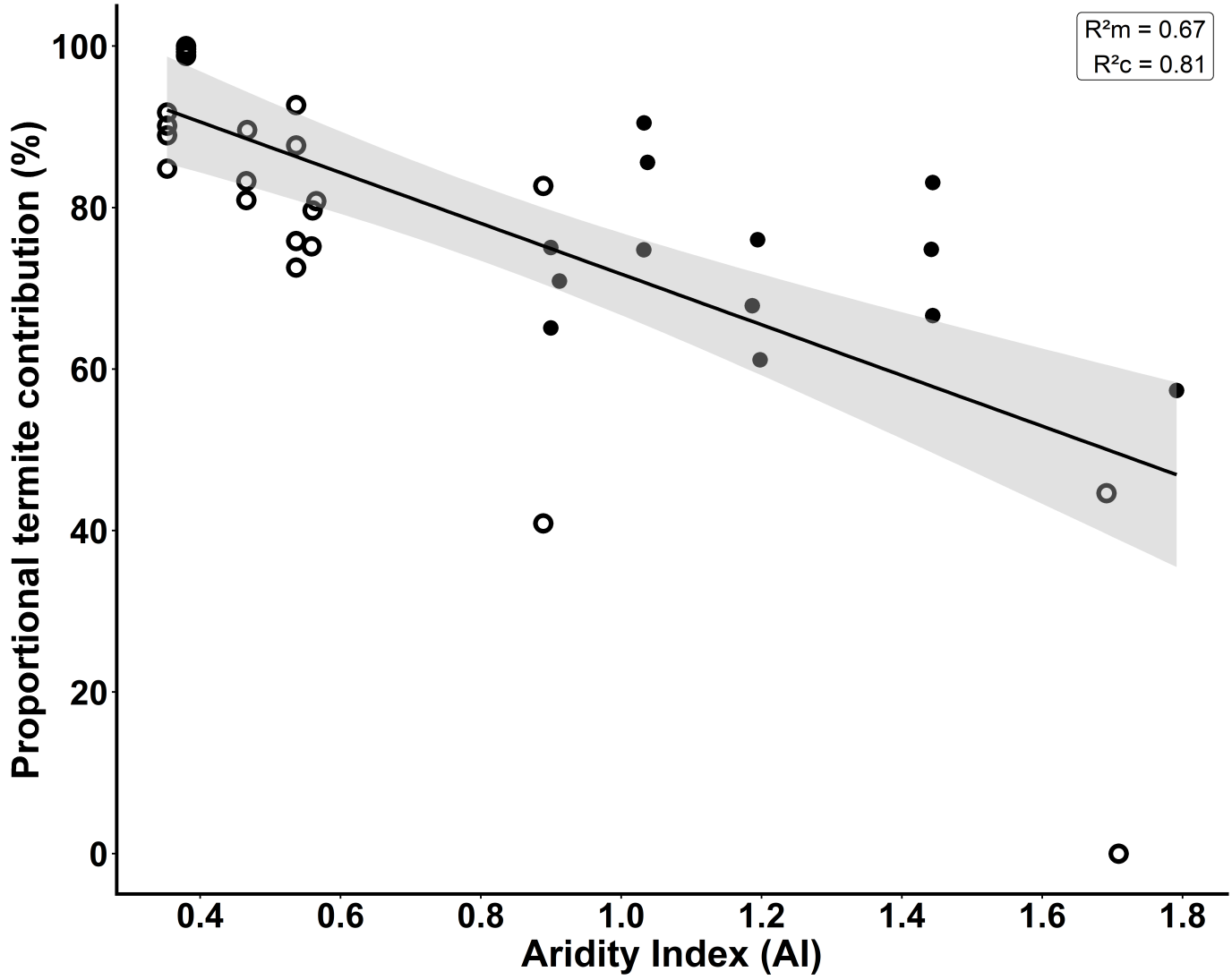
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
105.08	-35.40	NA	NA	NA	NA	4	-138.80	297.95	8.83	0.01
382.12	-44.32	-9.96	+	NA	+	7	-127.54	298.31	9.20	0.01
119.32	-34.73	-0.59	NA	NA	NA	5	-136.99	300.82	11.71	0.00
69.81	NA	NA	NA	NA	NA	3	-146.99	306.51	17.39	0.00
74.26	NA	NA	+	NA	NA	4	-143.59	307.91	18.80	0.00
156.97	NA	-3.46	NA	NA	NA	4	-144.33	308.04	18.93	0.00
165.17	NA	-3.60	+	NA	NA	5	-140.89	309.49	20.37	0.00
-53.68	NA	4.82	+	NA	+	6	-136.88	312.23	23.11	0.00

Selected_model_formula
Term_Prop ~ AI + VegType + (1 | Site) + AI:VegType

nobs	AIC	BIC	logLik	sigma
37	265.07	274.74	-126.54	8.74

term	estimate	std_error	df	t_value
(Intercept)	89.78	16.63	14.47	5.40
AI	-14.64	13.52	13.32	-1.08
VegTypeSavanna	21.67	17.45	14.90	1.24
AI:VegTypeSavanna	-37.86	15.57	11.90	-2.43

R2m	R2c
0.67	0.81



5 5) Microbe proportional (Microbe_Prop)

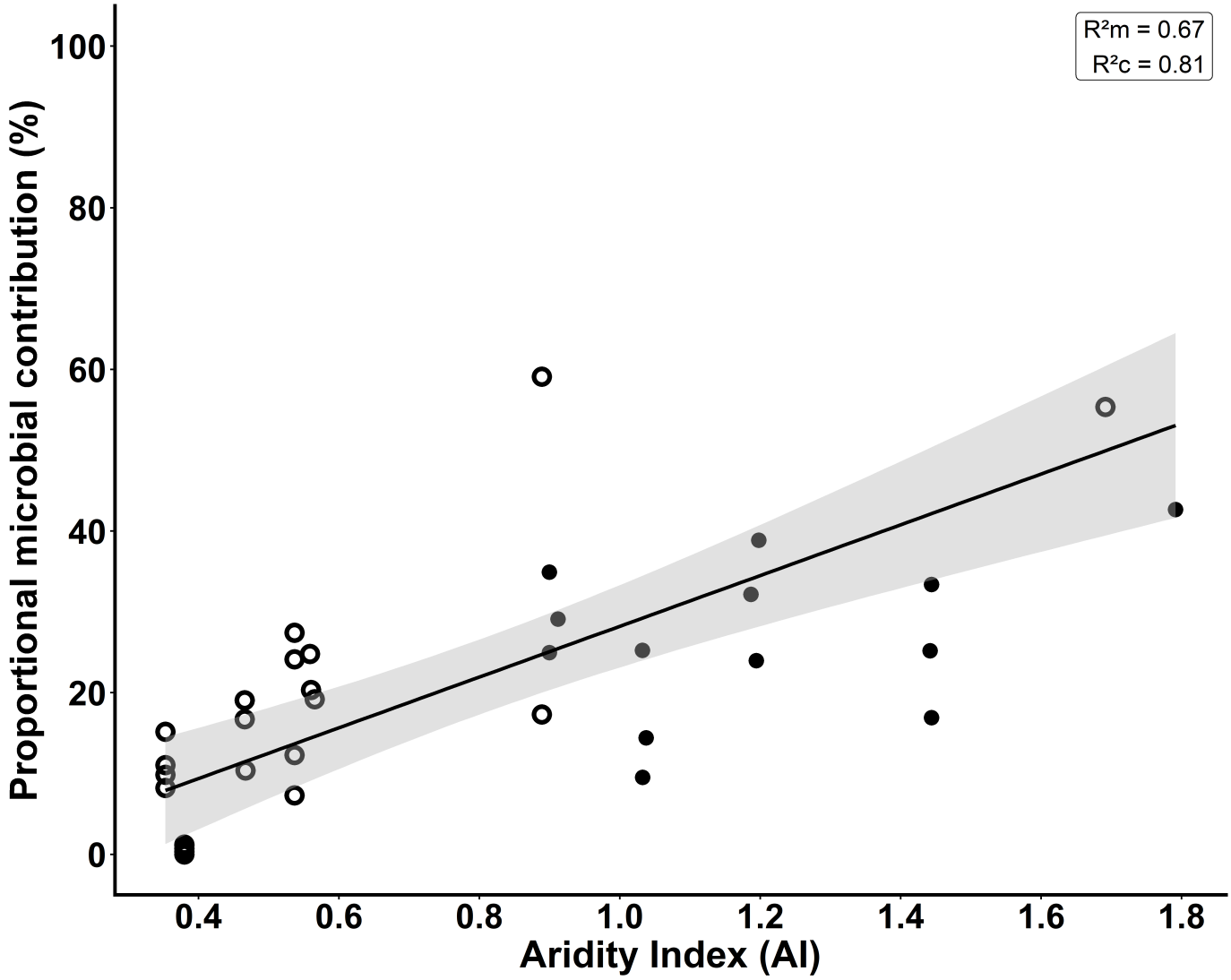
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
10.22	14.64	NA	+	+	NA	6	-126.54	289.11	0.00	0.69
0.32	14.55	0.40	+	+	NA	7	-125.27	292.56	3.45	0.12
-23.61	43.37	NA	+	NA	NA	5	-132.77	292.57	3.46	0.12
-41.33	42.21	0.76	+	NA	NA	6	-131.23	295.47	6.36	0.03
85.52	11.80	-2.76	+	+	+	8	-121.86	296.41	7.30	0.02
-5.08	35.40	NA	NA	NA	NA	4	-138.80	297.95	8.83	0.01
-282.12	44.32	9.96	+	NA	+	7	-127.54	298.31	9.20	0.01
-19.32	34.73	0.59	NA	NA	NA	5	-136.99	300.82	11.71	0.00
30.19	NA	NA	NA	NA	NA	3	-146.99	306.51	17.39	0.00
25.74	NA	NA	+	NA	NA	4	-143.59	307.91	18.80	0.00
-56.97	NA	3.46	NA	NA	NA	4	-144.33	308.04	18.93	0.00
-65.17	NA	3.60	+	NA	NA	5	-140.89	309.49	20.37	0.00
153.68	NA	-4.82	+	NA	+	6	-136.88	312.23	23.11	0.00

Selected_model_formula
Microbe_Prop ~ AI + VegType + (1 | Site) + AI:VegType

nobs	AIC	BIC	logLik	sigma
37	265.07	274.74	-126.54	8.74

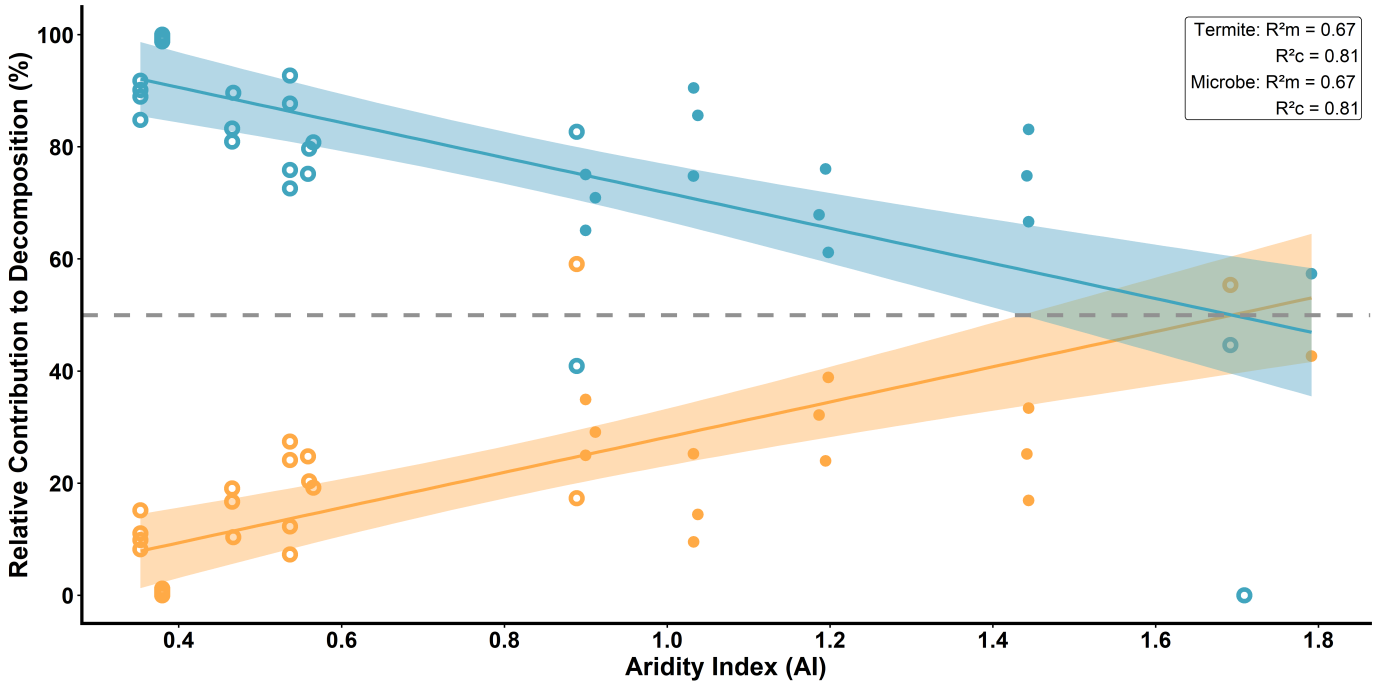
term	estimate	std_error	df	t_value
(Intercept)	10.22	16.63	14.47	0.61
AI	14.64	13.52	13.32	1.08
VegTypeSavanna	-21.67	17.45	14.90	-1.24
AI:VegTypeSavanna	37.86	15.57	11.90	2.43

R2m	R2c
0.67	0.81



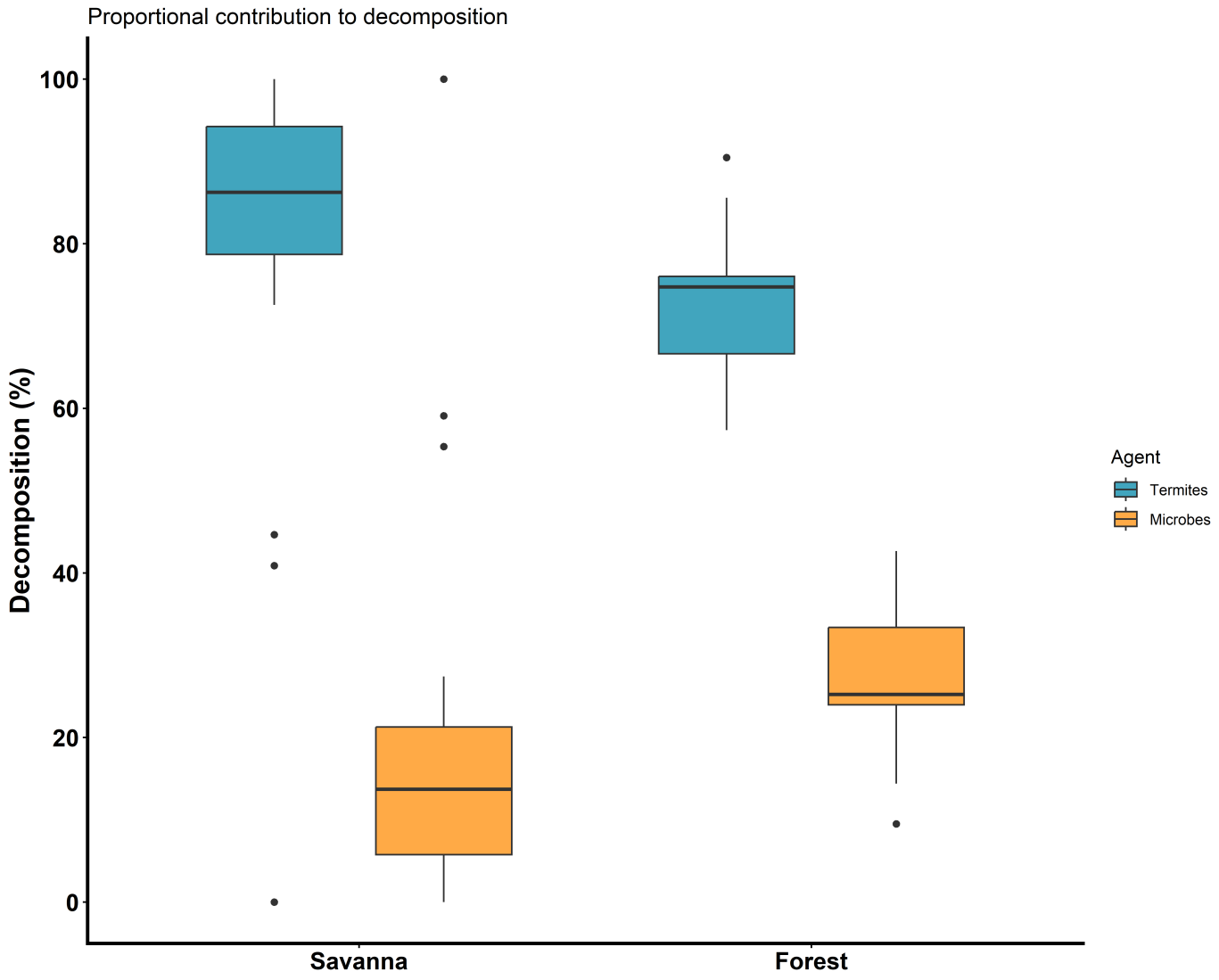
6 6) Combined proportional plot (termite +

microbe)

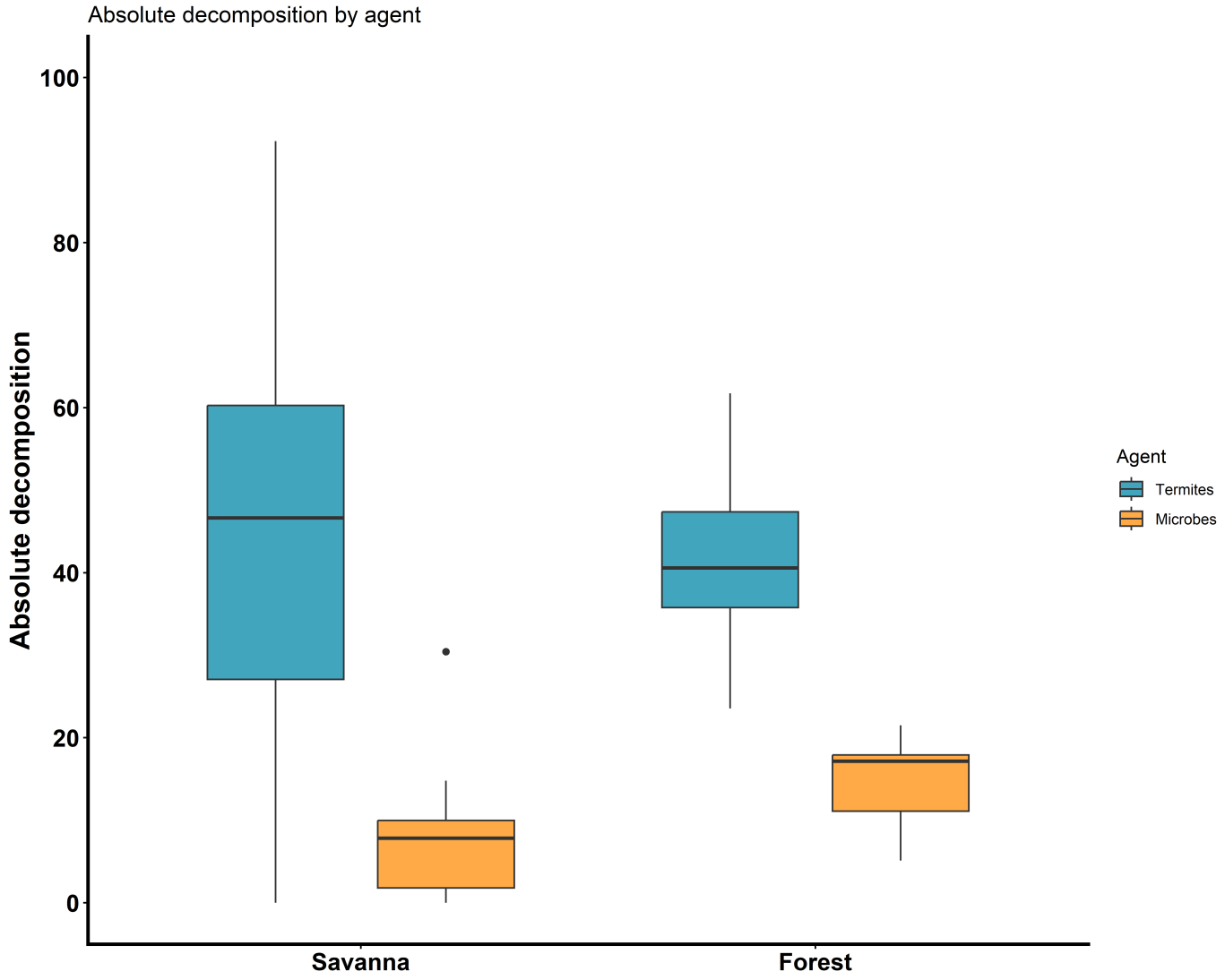

Termite: R²m = 0.67
R²c = 0.81
Microbe: R²m = 0.67
R²c = 0.81
Relative Contribution to Decomposition (%)
100
80
60
40
20
0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Aridity Index (AI)

7 7) Boxplots (no points; filled by agent)

7.1 Proportional contributions (0–100)



7.2 Absolute decomposition (0–100)



7.3 Two-way ANOVA: absolute decomposition by

habitat type and decomposition agent

```
library(dplyr)
library(tidyr)
library(car)
library(knitr)

Final_DB <- Final_DB %>%
  mutate(
    VegType = as.factor(VegType)
  )

abs_decomp_long <- Final_DB %>%
  select(VegType, Term_Absolute, Microbe_Absolute) %>%
  pivot_longer(
    cols = c(Term_Absolute, Microbe_Absolute),
    names_to = "Decomposition_type",
    values_to = "Absolute_decomposition"
  ) %>%
  mutate(
    Decomposition_type = factor(
      Decomposition_type,
      levels = c("Term_Absolute", "Microbe_Absolute"),
      labels = c("Termite", "Microbial")
    )
  ) %>%
  filter(
    !is.na(VegType),
    !is.na(Decomposition_type),
    !is.na(Absolute_decomposition)
  )

abs_decomp_aov <- aov(
  Absolute_decomposition ~ VegType * Decomposition_type,
  data = abs_decomp_long
)

abs_decomp_anova <- as.data.frame(car::Anova(abs_decomp_aov, type = 2)) %>%
  tibble::rownames_to_column("term")

knitr::kable(
  abs_decomp_anova,
  digits = 4,
  caption = "Two-way ANOVA for absolute decomposition by habitat type and decomposition agen
t"
)
```

Two-way ANOVA for absolute decomposition by habitat type and decomposition agent

term	Sum Sq	Df	F value	Pr(>F)
VegType	57.4373	1	0.2929	0.5901
Decomposition_type	20808.5515	1	106.1130	0.0000
VegType:Decomposition_type	493.2020	1	2.5151	0.1173

term	Sum Sq	Df	F value	Pr(>F)
Residuals	13726.8584	70	NA	NA

7.4 Mean absolute decomposition by habitat type

```
abs_decomp_summary <- abs_decomp_long %>%
  group_by(VegType, Decomposition_type) %>%
  summarise(
    n = n(),
    mean = mean(Absolute_decomposition, na.rm = TRUE),
    sd = sd(Absolute_decomposition, na.rm = TRUE),
    se = sd / sqrt(n),
    .groups = "drop"
  )

knitr::kable(
  abs_decomp_summary,
  digits = 3,
  caption = "Mean absolute decomposition by habitat type and decomposition agent"
)
```

Mean absolute decomposition by habitat type and decomposition agent

VegType	Decomposition_type	n	mean	sd	se
Forest	Termite	13	41.443	9.610	2.665
Forest	Microbial	13	14.921	4.809	1.334
Savanna	Termite	24	45.005	22.175	4.526
Savanna	Microbial	24	7.668	6.696	1.367

7.5 Overall mean absolute decomposition by decomposition agent

```
overall_abs_decomp_summary <- abs_decomp_long %>%
  group_by(Decomposition_type) %>%
  summarise(
    n = n(),
    mean = mean(Absolute_decomposition, na.rm = TRUE),
    sd = sd(Absolute_decomposition, na.rm = TRUE),
    se = sd / sqrt(n),
    .groups = "drop"
  )

knitr::kable(
  overall_abs_decomp_summary,
  digits = 3,
  caption = "Overall mean absolute decomposition for termite and microbial decomposition"
)
```

Overall mean absolute decomposition for termite and microbial decomposition

Decomposition_type	n	mean	sd	se
Termite	37	43.754	18.653	3.066
Microbial	37	10.216	6.977	1.147